

OmniConnect Blueprint

This document serves as the structural foundation for the Maton-inspired AI Gateway project.

FILE: README.MD

OmniConnect (Project-X)

The Universal Execution Layer for AI Agents in China

OmniConnect is a high-performance AI Gateway and Tool Orchestrator designed to bridge the gap between LLMs (Large Language Models) and the Chinese software ecosystem. Inspired by the "Execution Layer" philosophy, it provides a unified interface for agents to interact with local platforms like Feishu, DingTalk, and WeChat Work via MCP (Model Context Protocol).

🚀 Vision

To become the "central nervous system" for AI agents, providing secure, authenticated, and standardized "hands" to manipulate existing APIs without the complexity of manual OAuth management.

⚙️ Core Features

- **Unified Auth Gateway:** Centralized OAuth2/Token management for major Chinese SaaS platforms.
- **MCP-Native:** Built from the ground up to support the Model Context Protocol for seamless tool discovery.
- **Rust-Based Core:** High-concurrency, memory-safe ingress/egress gateway (leveraging experience from Panda architecture).
- **Privacy-First:** Support for local-first deployment and PII (Personally Identifiable Information) filtering.
- **Hybrid Routing:** Intelligently route tasks between local inference (llama.cpp) and cloud APIs.

🏗️ Tech Stack

- **Gateway:** Rust (Axum/Tokio)
- **Plugin System:** Python/TypeScript (MCP Servers)
- **Database:** PostgreSQL (for token storage) + Redis (for caching)
- **Deployment:** Docker / K8s

📁 Project Structure

- `/gateway``: The Rust-based proxy and auth manager.

- `/mcp-servers`: Standardized connectors for Feishu, DingTalk, etc.
- `/dashboard`: Management UI for API keys and tool monitoring.

Project Roadmap: OmniConnect

Phase 1: Foundation (Weeks 1-4)

- [] **Core Gateway Architecture:** Implement the basic Rust-based proxy layer.
- [] **Auth Engine:** Build the centralized OAuth2 handler for Feishu (Lark).
- [] **MCP Integration:** Setup the base MCP server structure to register tools.
- [] **Basic Security:** Implement API key management for agent access.

Phase 2: Tool Expansion (Weeks 5-8)

- [] **Enterprise Connectors:**
 - [] Feishu (Bitable, Calendar, Messaging)
 - [] DingTalk (Workflow, Messaging)
 - [] WeChat Work (Customer Management)
- [] **Schema Registry:** A discovery service for LLMs to understand available tool capabilities.
- [] **Documentation:** Generate OpenAPI/MCP specs automatically.

Phase 3: Advanced Features & Compliance (Weeks 9-12)

- [] **Data Localization:** Ensure all token/log storage complies with PIPL regulations.
- [] **Content Moderation:** Integrated filtering for upstream/downstream AI content.
- [] **Hybrid Inference:** Integration with local LLM providers for sensitive tasks.

Phase 4: Scaling & Ecosystem (Ongoing)

- [] **Developer SDK:** Simplify connecting new agents to OmniConnect.
- [] **Skill Marketplace:** Allow community-contributed MCP servers.
- [] **Monitoring:** Real-time observability dashboard for API usage and costs.